

$$\Rightarrow x = 9824 - 8074 \quad \Rightarrow x = 1750.$$

$\begin{array}{r} 9824 \\ - 8074 \\ \hline 1750 \end{array}$	$\begin{array}{r} 8888 \\ 848 \\ + 88 \\ \hline 9824 \end{array}$	$\begin{array}{r} 7337 \\ + 737 \\ \hline 8074 \end{array}$
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91. Let $414 \times x \times 7 = 127512$. Then

$$x = \frac{127512}{414 \times 7} = \frac{18216}{414} = \frac{2024}{46} = \frac{1012}{23} = 44.$$

$$\begin{array}{r} 44 \\ 23 \overline{)1012} \\ \underline{92} \\ 92 \\ \underline{0} \\ \end{array}$$

92. $82540027 \times 43253 = 82540027 \times (40000 + 3000 + 200 + 50 + 3)$
 $= (82540027 \times 40000) + (82540027 \times 3000) + (82540027 \times 200)$
 $+ (82540027 \times 50) + (82540027 \times 3)$

$$\begin{array}{r} 3301601080000 \\ 247620081000 \\ 16508005400 \\ 4127001350 \\ + 247620081 \\ \hline 3570103787831 \end{array}$$

Shortcut Method:

Product of unit's digits of the given numbers $= 7 \times 3 = 21$
 Clearly, the required product will have 1 as the unit's digit, which is 3570103787831.

93. Let $\frac{46351 - 36418 - 4505}{x} = 1357$. Then

$$x = \frac{46351 - (36418 + 4505)}{1357}$$

$$\Rightarrow x = \frac{(46351 - 40923)}{1357} = \frac{5428}{1357} = 4.$$

$\begin{array}{r} 46351 \\ - 40923 \\ \hline 5428 \end{array}$	$\begin{array}{r} 36418 \\ + 4505 \\ \hline 40923 \end{array}$
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94. $6 \times 66 \times 666 = 6 \times (6 \times 11) \times (6 \times 111)$
 $= (6 \times 6 \times 6) \times (11 \times 111) = (216 \times 1221)$
 $= (1221 \times 216) = 1221 \times (200 + 10 + 6)$
 $= (1221 \times 200) + (1221 \times 10) + (1221 \times 6)$
 $= 244200 + 12210 + 7326 = 263736.$

$$\begin{array}{r} 244200 \\ 12210 \\ + 7326 \\ \hline 263736 \end{array}$$

95. $(+1) - (-1) = (+1 + 1) = 2.$

96.

$$\begin{array}{r} 888888 \\ 88888 \\ 8888 \\ 888 \\ 88 \\ + 8 \\ \hline 987648 \end{array}$$

97. Given Expression $= 17 + (-12) - 48 = 17 - 60 = -43.$

98. $\frac{60840}{234} = \frac{30420}{117} = \frac{3380}{13} = 260.$

99. Let $3578 + 5729 - x \times 581 = 5821$. Then

$$x \times 581 = (3578 + 5729) - 5821 \Rightarrow x \times 581 = (9307 - 5821)$$

$$= 3486 \Rightarrow x = \frac{3486}{581} = 6.$$

100. $-95 \div 19 = \frac{-95}{19} = -5.$

101. $12345679 \times 72 = 12345679 \times (70 + 2)$
 $= (12345679 \times 70) + (12345679 \times 2)$
 $= (864197530 + 24691358)$
 $= 888888888.$

$$\begin{array}{r} 864197530 \\ + 24691358 \\ \hline 888888888 \end{array}$$

102. $8899 - 6644 - 3322 = x - 1122 \Rightarrow 2255 - 3322 + 1122 = x$
 $\Rightarrow x = 3377 - 3322 = 55.$

103. Let $74844 \div x = 54 \times 63$. Then, $\frac{74844}{x} = 54 \times 63$
 $\Rightarrow x = \frac{74844}{54 \times 63} = \frac{8316}{6 \times 63} = \frac{1386}{63} = \frac{198}{9} = 22.$

104. $1256 \times 3892 = (1000 + 200 + 50 + 6) \times 3892$
 $= (1000 \times 3892) + (200 \times 3892) +$
 $(50 \times 3892) + (6 \times 3892)$
 $= 3892000 + 778400 + 194600 + 23352$
 $= 4888352.$

$$\begin{array}{r} 3892000 \\ 778400 \\ 194600 \\ + 23352 \\ \hline 4888352 \end{array}$$

105. $786 \times 964 = (800 - 14) \times 964$
 $= (800 \times 964) - (14 \times 964)$
 $= (771200 - 13496) = 757704.$

$$\begin{array}{r} 771200 \\ - 13496 \\ \hline 757704 \end{array}$$

106. $348 \times 265 = (350 - 2) \times 265 = (350 \times 265) - (2 \times 265)$
 $= \{(300 + 50) \times 265\} - 530 = (300 \times 265)$
 $+ (50 \times 265) - 530$
 $= 79500 + 13250 - 530 = 92750 - 530$
 $= 92220.$

107. $(71 \times 29 + 27 \times 15 + 8 \times 4) = (80 - 9) \times 29 + 405 + 32 =$
 $(80 \times 29) - (9 \times 29) + 437 = 2320 - 261 + 437 = 2757 - 261$
 $= 2496.$

108. Let $x \times (|a| \times |b|) = -ab$. Then, $x = \frac{-ab}{|ab|} = -1.$

109. Let $(46)^2 - x^2 = 4398 - 3066$
 Then, $(46)^2 - x^2 = 1332 \Rightarrow x^2 = (46)^2 - 1332$
 $\therefore x^2 = (50 - 4)^2 - 1332 = (50)^2 + 4^2 - 2 \times 50 \times 4 - 1332$
 $\Rightarrow x^2 = 2500 + 16 - 400 - 1332 = 2516 - 1732 = 784$
 $\Rightarrow x = \sqrt{784} = 28.$

$\begin{array}{r} 2 \overline{)784} \quad (28 \\ 4 \\ \underline{4} \\ 384 \\ 384 \\ \underline{0} \\ \end{array}$	$\begin{array}{r} 2516 \\ - 1732 \\ \hline 784 \end{array}$
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110. $(800 \div 64) \times (1296 \div 36) = \frac{800}{64} \times \frac{1296}{36} = 450.$